A D D E N D U M

Q A P R O C E D U R E

GE 3.0T High Resolution Brain Array

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	MRIDC #	Qty
18cm Diameter Spherical TLT Phantom w Silicon Oil	2359877	N/A	1
Phantom Positioner	2317112-3	102558	1

3-2-2 Data Collection

1. Attach the base plate with 8 Channel HR Brain coil to the cradle in the same position on the cradle as the standard head coil. Note the serial number of the coil.

2. Slide the coil back to permit access to the head holder as shown in Figure 3-2-2-1.

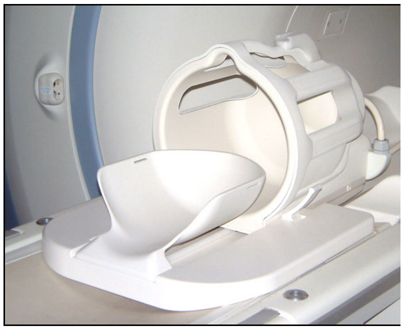
3. Place phantom ring on head holder as shown in Figure 3-2-2-2. The edge of the ring that goes toward the magnet has edges that overhang the head holder to secure the ring in position.

4. Place head TLT sphere on phantom ring as shown in Figure 3-2-2-3.

5. Slide the coil all the way forward (away from the magnet) on the base plate and set the locks as shown in Figure 3-2-2-4.

6. Establish an axial landmark on the line on top of the coil. Advance to scan.

Figure 3-2-2-1



Slide Coil Back to Access Head Holder

Figure 3-2-2-2

Place Phantom Ring on Head Holder

Figure 3-2-2-3

Place Head TLT Sphere on Phantom Ring

Figure 3-2-2-4

Sphere Aligned with Coil

7. End the previous exam if this has not already been done.

8. On the service desktop, open a command window and navigate to the **/export/home/signa/tools/CoilsTools** directory.

9. Enter the command **run_multicoil_QA** and press the enter key.

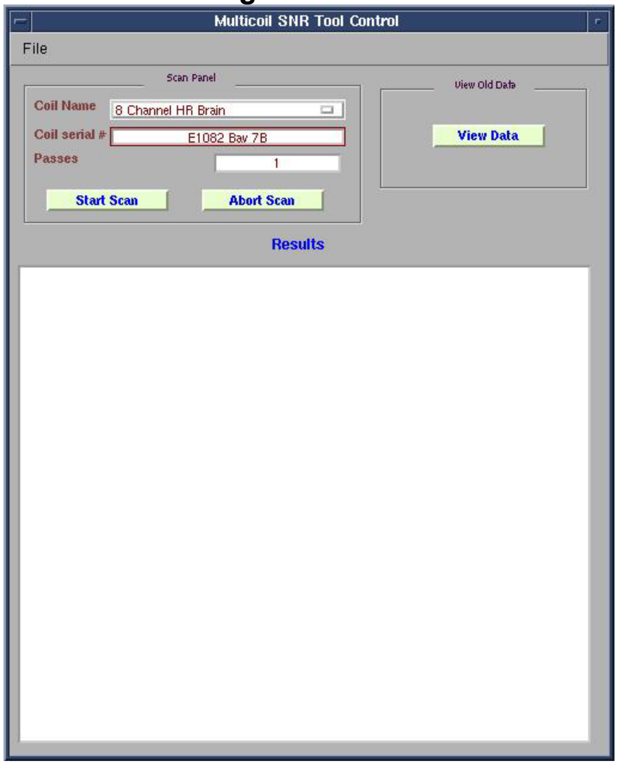
10. When the Multicoil SNR Tool Control window (Figure 3-2-2-5) opens, select the 8 Channel HR Brain from the Coil Name selection menu.

11. Enter the Coil serial number.

12. Enter the number of repetitions in the Passes field.

13. Hit the Start Scan button.

Figure 3-2-2-5



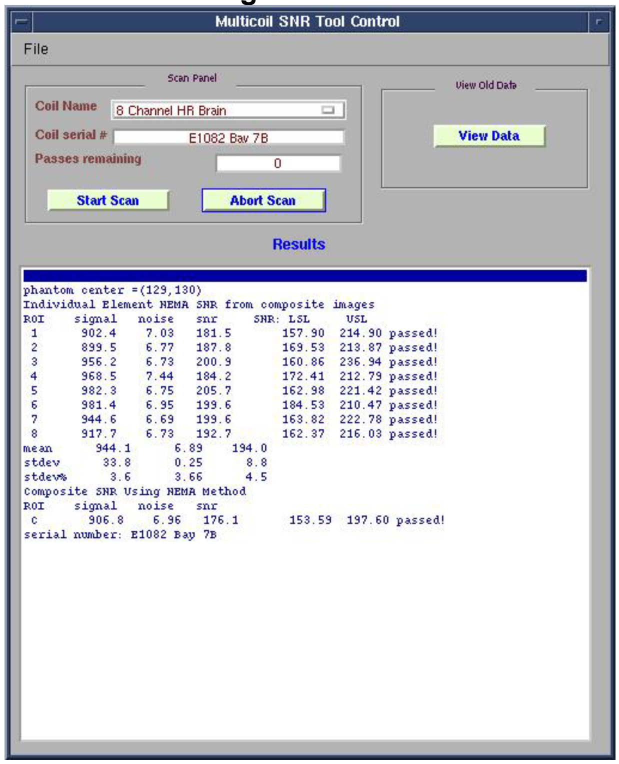
Multicoil SNR Tool Control window

3-2-3 Data Analysis

The data analysis is automated.

1. After the scan is completed, the data will be displayed in the Multicoil SNR Tool Control window (Figure 3-2-3-1).

Figure 3-2-3-1



Results displayed in the Multicoil SNR Tool Control window

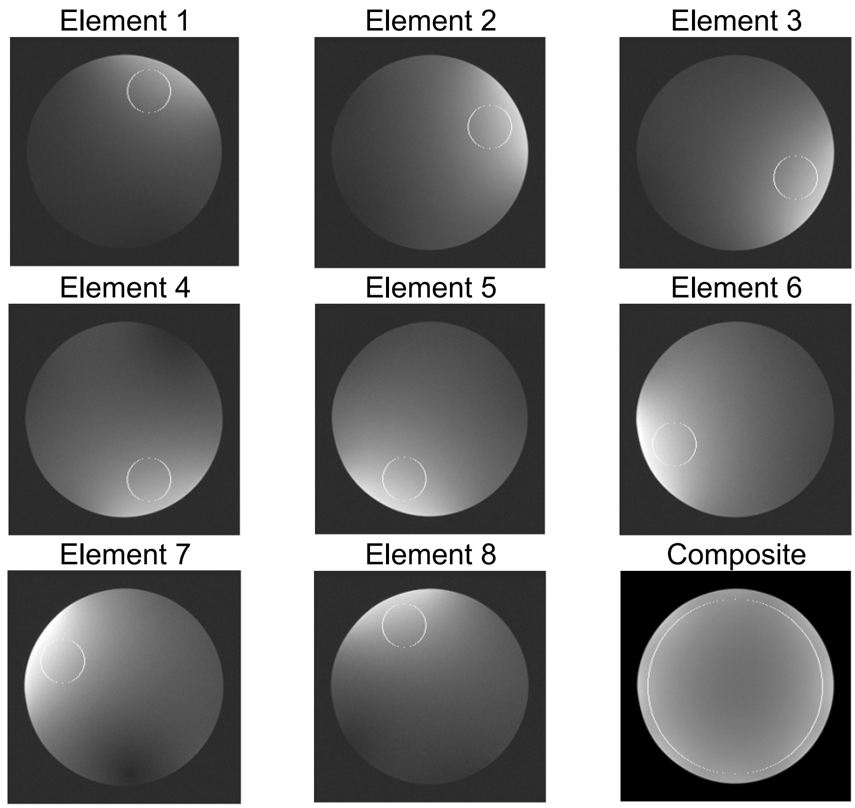
COMPOSITE SNR ACCEPTANCE LIMITS – TABLE 3-2-3-2

Parameter	MEAN	STDEV	%Mean	USL	LSL
Composite SNR	175.62	7.34	4.18	202	149

INDIVIDUAL ELEMENT SNR LIMITS - TABLE 3-2-3-3

Parameter	MEAN	STDEV	%Mean	USL	LSL
Ch 1 Ind. Element - Composite	186.42	9.50	5.10	214	158
Ch 2 Ind. Element - Composite	191.68	7.39	3.86	220	163
Ch 3 Ind. Element - Composite	189.92	12.69	6.68	218	161
Ch 4 Ind. Element - Composite	192.63	6.73	3.49	222	164
Ch 5 Ind. Element - Composite	192.20	9.74	5.07	221	163
Ch 6 Ind. Element - Composite	197.57	4.32	2.19	227	168
Ch 7 Ind. Element - Composite	193.32	9.83	5.08	222	164
Ch 8 Ind. Element - Composite	189.22	8.94	4.73	218	161

Figure 3-2-3-4

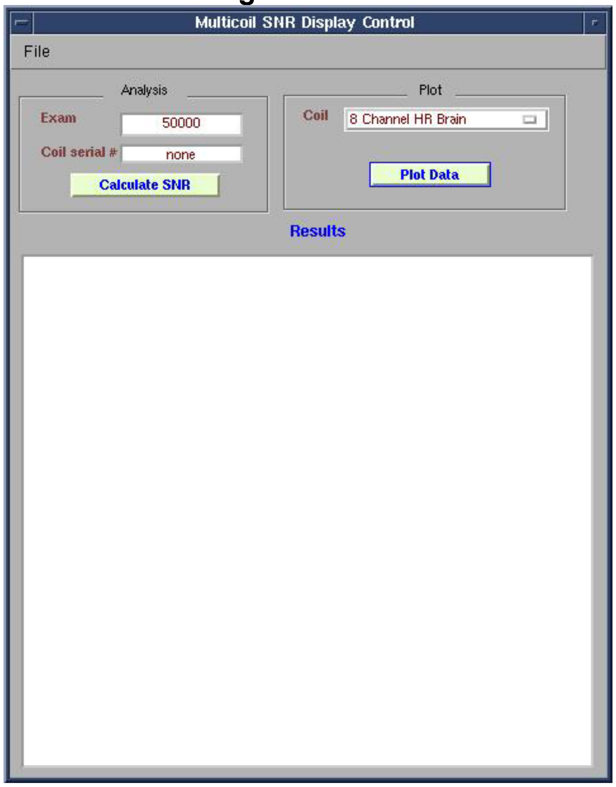


Individual Element ROIs

To view archived data files follow:

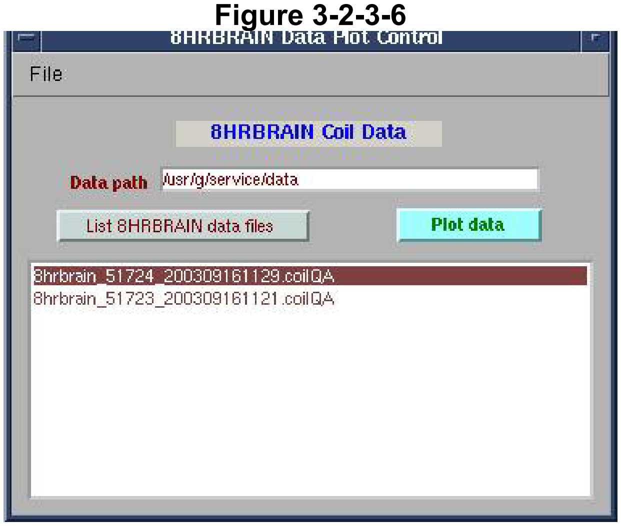
2. Click on the View Data button in the Multicoil SNR Tool Control window (Figure 3-2-3-1).
3. Select the File Tab in the now open Multicoil SNR Display Control window (figure 3-2-3-5).
4. Turn Expert Mode on
5. Select the 8 Channel HR Brain coil in the coil selection menu.
6. Click the Plot Data button.
7. The Data Plot Control window opens (Figure 3-2-3-6).

Figure 3-2-3-5



Multicoil SNR Display Control window

Figure 3-2-3-6

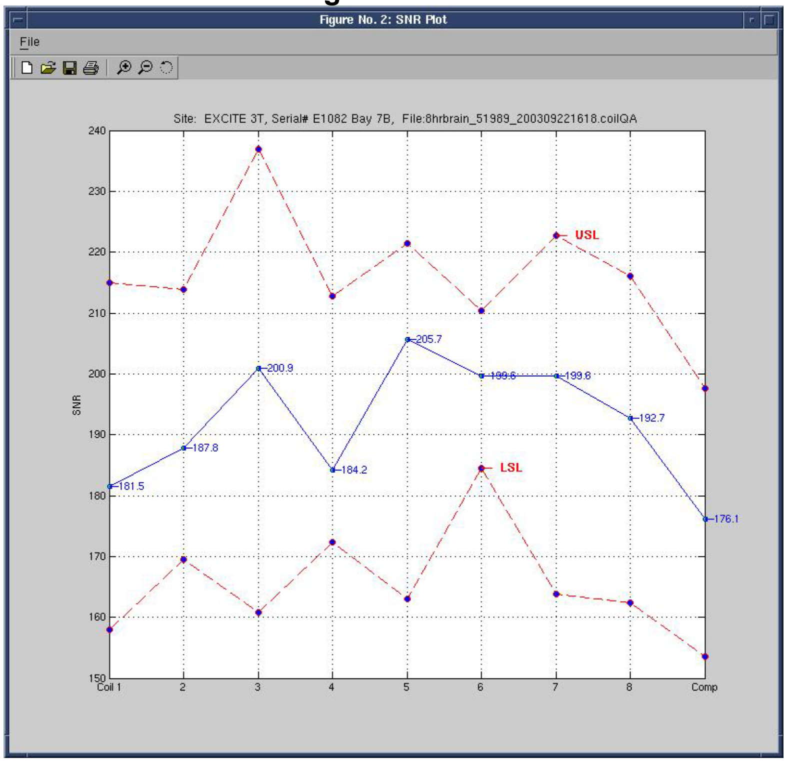


Data Plot Control window

8. Select the desired file and click Plot Data

9. A plot window with SNR, Signal and Noise plots will open (Figure 3-2-3-7).

Figure 3-2-3-7



Nominal Individual Element Results

3-2-4 Test Explanation and Results Interpretation

The 8 Channel HR Brain coil is a moderately complex phased array. The Quality Assurance tool acquires data looking at the composite image (like the customer sees). Individual element performance is evaluated using the composite image.

Signal images are acquired using a FSE-XL sequence with CV saveinter set to 1 to save intermediate images into the image database. The scan is taken twice for a total of 18 images. Next, a no-RF noise scan is taken, also with saveinter on for a total of 9 noise images. Individual element signal and noise images no longer yield results that are displayed to the user.

The test employs two types of image analyses: Separate signal and noise images are evaluated for individual element signal, noise and SNR. In addition, a filtered version of the NEMA SNR measurement method is also used. The filter takes out adverse effects of subtle ghosting on noise measured in difference.